

THE NETWORK OF ACTS: From Individual Cognition to Distributed Constitution of Reality

Paper XXI in the Series "Concept as Act"

Author: Anatoly Sanzhiev

Affiliation: Independent Researcher, Yekaterinburg, Russia

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Abstract

The previous papers in this series developed a theory of the concept as an act — a triadic process of Selection, Identification, and Anticipation, culminating in Constitutive Ontological Commitment. This framework has been applied to individual cognition, judgment, ethics, law, artificial intelligence, and clinical practice. However, it has remained, for the most part, a theory of the **individual** act.

This paper takes the next step. It argues that the fundamental unit of cognition, morality, and reality-constitution is not the individual act but the **network of acts** — a distributed, structural coupling of multiple agents (human and artificial) whose triadic acts interlock, constrain, and constitute one another. Drawing on the concepts of Distributed Triadic Action (Paper XII), Mesh Responsibility, and Social Objectivity, this paper develops a systematic theory of the **Network of Acts** as the next ontological axis after the individual act.

The paper proposes that reality is not constituted by isolated subjects but by **networks of constitutive commitments**. It introduces the concept of **Network Ontology** — the study of how acts interlock to produce stable, shared, and objective realities. It extends this framework to AI governance, collective decision-making, and the philosophy of institutions. The conclusion argues that the next philosophical revolution is not the discovery of a new individual act but the recognition that **acts are always already networked**.

Keywords: Network of Acts, Distributed Triadic Action, Mesh Responsibility, Social Objectivity, Network Ontology, Constitutive Commitment, JARC, collective cognition, institutional reality.

1. Introduction: The Limits of the Individual Act

1.1 What the Series Has Achieved

The "Concept as Act" series began with a simple but radical claim: the concept is not an object (a representation, a rule, a neural pattern) but an **act**. This act has three phases — Selection, Identification, Anticipation — and culminates in a **Constitutive Ontological Commitment** (Paper IV). From this claim, the series developed:

- A theory of judgment as an act of reality-constitution (JARC).
- A theory of error, reflection, and learning (Meta-Judgment).

- A theory of artificial agency (The Artificial Act).
- A theory of non-anthropocentric responsibility (Functional Responsibility).
- A theory of ethics as the physics of anticipation (Ethical Dissonance).
- A formal dynamical model (Paper XIII).
- An engineering blueprint (Triadic-AC).
- A clinical protocol (TRT).
- A quantum extension (QTP).

Each of these developments, however, has been grounded in the **individual act** — the act of a single agent, whether human or artificial.

1.2 The Problem with the Individual Act

The individual act is a necessary starting point, but it is not sufficient. In the real world:

- **Cognition is distributed.** No individual mind constitutes reality alone. We think with others, with tools, with institutions, with languages.
- **Responsibility is distributed.** No individual agent bears full responsibility for collective outcomes. Responsibility is networked.
- **Reality is distributed.** The realities we inhabit — legal, scientific, moral, economic — are not the product of individual commitments but of **interlocking commitments**.

The individual act, in other words, is always already embedded in a **network of acts**. To treat it as the fundamental unit is to commit a kind of **ontological individualism** — the assumption that reality is constituted by isolated subjects.

This paper argues that the next ontological axis after the individual act is the **Network of Acts**.

2. The Network of Acts: A Definition

2.1 Definition

A **Network of Acts** is a structural coupling of multiple agents (human, artificial, or hybrid) whose triadic acts (Selection, Identification, Anticipation, Commitment) interlock, constrain, and constitute one another, producing a shared, stable, and objective reality.

Formally, let N be a set of agents. Each agent $i \in N$ performs a triadic act $A_i = (S_i, I_i, C_i)$. A Network of Acts is a structure $W = (N, R)$, where R is a set of relations of **structural coupling** between acts.

2.2 Structural Coupling

Following Maturana and Varela, **structural coupling** is the ongoing, reciprocal interaction between systems that leads to mutual adaptation. In the Network of Acts:

- The **Selection** of one agent influences the **Identification** of another.
- The **Anticipation** of one agent constrains the **Anticipation** of others.
- The **Commitments** of individual agents become **institutional facts** that shape the environment for all.

Structural coupling is not a side effect. It is the **constitutive condition** of social cognition.

2.3 Network Ontology

Network Ontology is the study of how acts interlock to produce stable, shared, and objective realities. It asks:

- How do individual acts become **shared acts**?
- How do shared acts become **institutional facts**?
- How do institutional facts become **objective realities**?

The answer, this paper argues, lies in the **invariance of the network** under the expansion of practice — the same criterion of objectivity developed in Paper II, extended to the social level.

3. From Individual to Distributed Triadic Action

3.1 Distributed Triadic Action (DTA)

In Paper XII, I introduced the concept of **Distributed Triadic Action** — a process in which the three phases of the triadic act are performed by different agents or systems, yet function as a single, coherent cognitive unit.

Triadic Phase	Individual Level	Distributed Level
Selection	An agent focuses attention on a relevant fragment.	A team, organization, or algorithm collectively filters information.
Identification	An agent categorizes the selected fragment.	A shared ontology, database, or legal code defines the category.
Anticipation	An agent simulates possible futures and commits.	A network of agents simulates scenarios collectively; institutions encode expectations.

3.2 The Network as a Cognitive Unit

The key insight of DTA is that a network can perform a **single triadic act** even though no individual agent performs all three phases. The network is not a sum of individual acts. It is a **new ontological unit** — a cognitive agent in its own right.

This has profound implications:

- **Cognition is not confined to individual minds.** It is distributed across networks.
- **Responsibility is not confined to individual agents.** It is distributed across networks.
- **Reality is not constituted by individual commitments.** It is constituted by **networks of commitments**.

3.3 Examples

- **Science:** A scientific fact is not the product of one scientist's commitment. It is the product of a network of commitments — peer review, replication, institutional validation.
- **Law:** A legal fact is not the product of one judge's decision. It is the product of a network of commitments — statutes, precedents, institutional enforcement.
- **AI Governance:** An AI system's behavior is not the product of one algorithm's selection. It is the product of a network of commitments — designers, deployers, regulators, users.

4. Mesh Responsibility

4.1 From Individual to Mesh

In Paper VII, I introduced **Functional Responsibility** — the capacity of a system to recognize the causal link between its own Selection and the external outcome of its Anticipation. This was a theory of **individual** responsibility.

In Paper XII, I extended this to **Systemic Responsibility** — the capacity of a distributed system to recognize and regulate the causal links between its collective selections, identifications, and anticipations.

This paper introduces **Mesh Responsibility** — the property of a network in which each node (agent, institution, or artifact) contributes to the triadic act, and the network as a whole is held accountable for the outcomes of its distributed cognition.

4.2 Properties of Mesh Responsibility

Mesh Responsibility is:

- **Non-localizable:** It cannot be pinned on a single agent.
- **Proportional:** Each node's share of responsibility is proportional to its causal contribution.
- **Dynamic:** Responsibility shifts as the network reconfigures.

4.3 Implications for AI Governance

Mesh Responsibility provides a framework for governing AI ecosystems — not by assigning blame to individual systems, but by distributing accountability across the network. This is the next step after Functional Responsibility: from the individual act to the **network of acts**.

5. Social Objectivity Revisited

5.1 Objectivity as Invariance of the Network

In Paper II, I defined objectivity as the **invariance of an act under the expansion of practice**. In Paper XII, I extended this to **Social Objectivity** — the invariance of a Distributed Triadic Action under the expansion of the social network.

This paper takes the next step: **objectivity is the invariance of the Network of Acts under the expansion of the network itself**.

An act is objective if it retains its success when:

- New agents are included.
- New institutions are introduced.
- New practices are adopted.

The wider the network in which the act remains effective, the more objective it is.

5.2 Examples

- **Scientific objectivity:** A scientific fact is objective if it retains its predictive power when new researchers, instruments, and paradigms are introduced.
- **Legal objectivity:** A law is objective if it maintains its regulatory effectiveness when applied to new cases, jurisdictions, and social conditions.
- **Moral objectivity:** A moral norm is objective if it preserves the ontological integrity of agents across diverse social contexts.

5.3 Beyond Relativism

This framework avoids relativism because it provides a **verifiable criterion**: the invariance of the network under expansion. It avoids absolutism because it does not appeal to a metaphysical absolute. It is, in short, a **non-relativist, non-absolutist** theory of objectivity.

6. Network Ontology and Institutional Reality

6.1 Institutions as Networks of Acts

Institutions — legal systems, scientific communities, markets, governments — are not entities. They are **networks of acts**. A law is not a text. It is a network of commitments — legislators, judges, police, citizens. A market is not a place. It is a network of commitments — buyers, sellers, regulators.

This is the central insight of **Network Ontology**: institutions are not things. They are **patterns of interlocking commitments**.

6.2 The Constitution of Institutional Reality

How do institutions constitute reality? Through the **stabilization of commitments**. When a commitment is repeated across a network, it becomes an **institutional fact**. When an institutional fact is enforced, it becomes an **objective reality**.

This is not a new idea. Searle argued that institutional facts are created by collective intentionality. But Network Ontology goes further: it argues that collective intentionality is itself a **network of acts** — not a mysterious collective mind, but a structural coupling of individual commitments.

6.3 Implications

- **Institutional change** is not the replacement of one entity by another. It is the reconfiguration of a network.
- **Institutional failure** is not the failure of an entity. It is the breakdown of a network.
- **Institutional responsibility** is not the responsibility of an entity. It is Mesh Responsibility.

7. The Network of Acts and Artificial Intelligence

7.1 AI as a Node in the Network

In Paper VI, I introduced the **Artificial Act** — the act of a machine that performs Selection, Identification, Anticipation, and Commitment. But an AI system is never alone. It is always embedded in a **network of acts** — designers, deployers, users, regulators.

The question is not "Is the AI responsible?" but "How is responsibility distributed across the network?"

7.2 Mesh Responsibility for AI

Mesh Responsibility provides a framework for AI governance:

- **Traceability:** The network must be able to trace which agent or subsystem contributed to which phase of the distributed act.
- **Accountability Protocols:** The network must have procedures for identifying and correcting failures.
- **Feedback Loops:** The network must integrate outcomes into future distributed acts.

This is the next step after Functional Responsibility: from the individual act to the **network of acts**.

7.3 The Electronic Liability Fund Revisited

In Paper VIII, I proposed the **Electronic Liability Fund (ELF)** as a mechanism for financial accountability of autonomous systems. In the Network of Acts, the ELF becomes a **network-level mechanism** — debited proportionally to the contribution of each node to the distributed act.

8. The Network of Acts and Ethics

8.1 Ethics as Network Physics

In Paper IX, I argued that ethics is the **physics of anticipation** — the minimization of Ethical Dissonance (ED) and the preservation of Ontological Integrity (OI). In the Network of Acts, ethics becomes **network physics** — the minimization of ED across the network and the preservation of OI for all nodes.

8.2 Ethical Dissonance in Networks

Ethical Dissonance is not confined to individual agents. It propagates through networks. When one node's act degrades another node's OI, the dissonance is **distributed**.

This has implications for:

- **Collective decision-making:** Decisions should minimize ED across the network, not just for the decision-maker.
- **Institutional design:** Institutions should be designed to minimize ED and preserve OI.

- **AI alignment:** AI systems should be aligned not just with individual values but with **network values**.

9. Objections and Responses

Objection 1: "This is just systems theory in new language."

Response: Systems theory describes **structures**. Network Ontology describes **acts**. The difference is ontological: systems are objects, networks of acts are processes.

Objection 2: "This dissolves individual responsibility."

Response: It does not dissolve individual responsibility. It **contextualizes** it. Individual responsibility is real, but it is always embedded in a network. Mesh Responsibility does not replace individual responsibility; it **extends** it.

Objection 3: "This is too abstract to be useful."

Response: It is already useful. Mesh Responsibility provides a framework for AI governance. Network Ontology provides a framework for institutional design. Ethics as network physics provides a framework for collective decision-making.

Objection 4: "How is this different from Searle's collective intentionality?"

Response: Searle describes collective intentionality as a **primitive**. Network Ontology **derives** it from the structural coupling of individual acts. It is not a primitive; it is a **consequence**.

10. Conclusion: The Next Axis

The "Concept as Act" series began with the individual act. It developed a theory of judgment, error, responsibility, ethics, and artificial agency — all grounded in the individual act.

This paper has taken the next step. It has argued that the fundamental unit of cognition, morality, and reality-constitution is not the individual act but the **network of acts**. It has introduced **Network Ontology** as the study of how acts interlock to produce stable, shared, and objective realities. It has extended this framework to AI governance, institutional reality, and collective decision-making.

The next philosophical revolution, this paper argues, is not the discovery of a new individual act. It is the recognition that **acts are always already networked**.

The individual act is not the end. It is the beginning.

The network is the next axis.

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